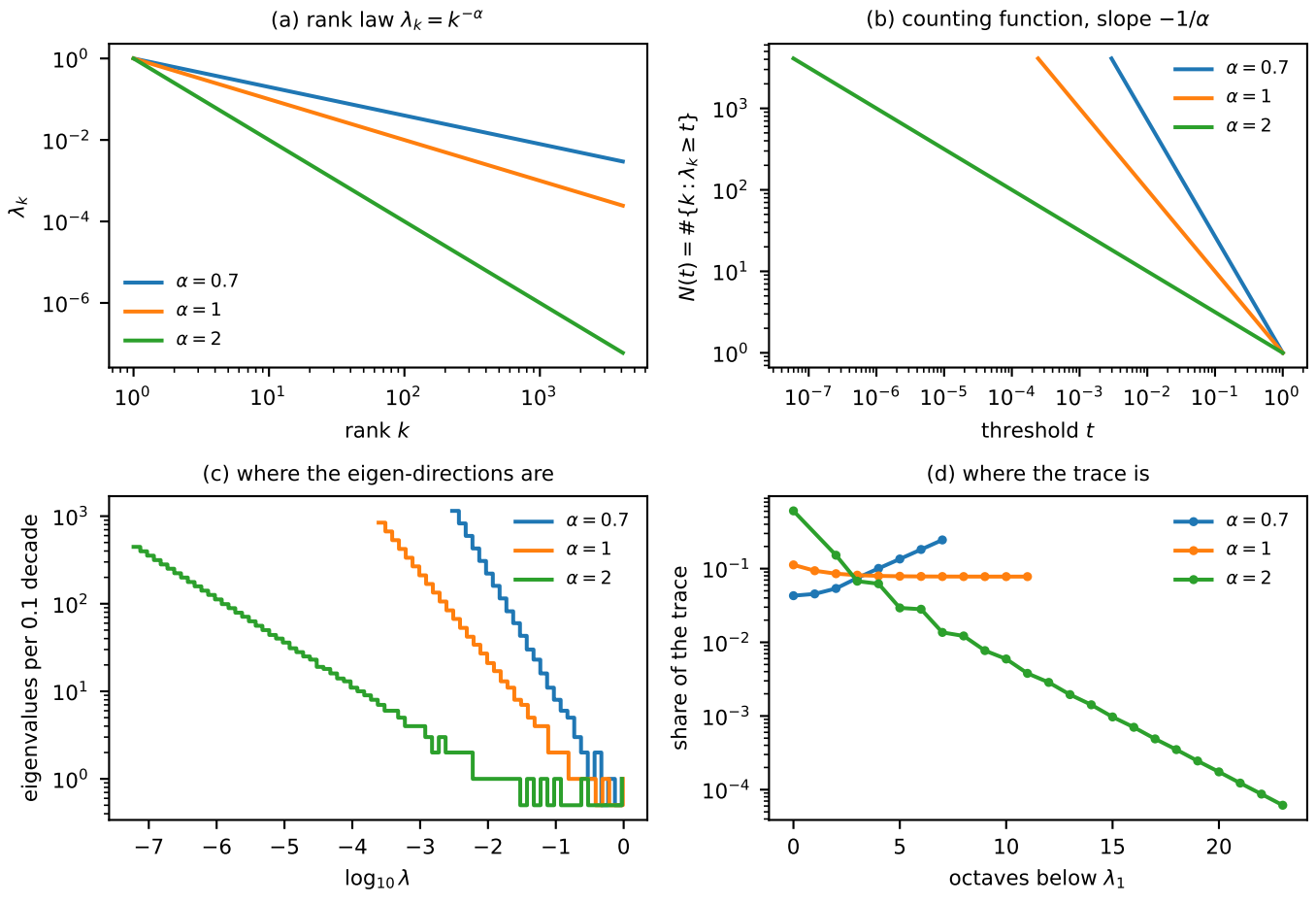
Four faces of a pure power law (D = 4096)
(a) rank law $\lambda_k = k^{-\alpha}$
λ_k
rank k
$\alpha = 0.7$
$\alpha = 1$
$\alpha = 2$
(b) counting function, slope $-1/\alpha$
$N(t) = \#\{k : \lambda_k \geq t\}$
threshold t
(c) where the eigen-directions are
eigenvalues per 0.1 decade
$\log_{10}\lambda$
(d) where the trace is
share of the trace
octaves below λ_1